Class XII Session 2025-26
Subject - Physics
Sample Question Paper - 3

Time Allowed: 30 Minutes

General Instructions:

1. There are 33 questions in all. All questions are compulsory.
2. This question paper has five sections: Section A,.
3. Use of calculators is not allowed.

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Section A

1. In forward bias the width of depletion layer **[1]**

   a) independent of V
   b) decreases with increase in V
   c) increases with increase in V
   d) dependent of V

2. According to Kirchhoff's Loop Rule, **[1]**

   a) The algebraic sum of changes in potential around any closed loop must be positive.
   b) The algebraic sum of changes in potential around any closed loop must be negative.
   c) The absolute sum of changes in potential around any closed loop must be zero.
   d) The algebraic sum of changes in potential around any closed loop must be zero.

3. If an object is placed unsymmetrically between two plane mirrors inclined at 70°, then the total number of images formed is **[1]**

   a) 4
   b) 5
   c) 1
   d) 3

4. The major contribution of magnetism in substances is due to **[1]**

   a) equally due to orbital and spin motions of electrons
   b) orbital motion of electrons

c) spin motion of electrons

d) hidden magnets.

5. A parallel plate capacitor has capacitance C. If it is equally filled with parallel layers of materials of dielectric constants K_1 and K_2, its capacity becomes C_1. The ratio of C_1 to C is **[1]**

a) $K_1 + K_2$

b) $\frac{K_1 K_2}{K_1 + K_2}$

c) $\frac{2K_1 K_2}{K_1 + K_2}$

d) $\frac{K_1 + K_2}{K_1 K_2}$

6. A thin ring of radius R metre has charge q coulomb uniformly spread on it. The ring rotates about its axis with a constant frequency of f revolutions/s. The value of magnetic induction in Wb/m^2 at the centre of the ring is: **[1]**

a) $\frac{\mu_0 q}{2\pi f R}$

b) $\frac{\mu_0 q f}{2R}$

c) $\frac{\mu_0 q f}{2\pi R}$

d) $\frac{\mu_0 q}{2fR}$

7. A square loop of wire of each side 50 cm is kept so that its plane makes an angle θ with a uniform magnetic field of induction IT. The magnetic field is withdrawn in 0.1 s. It is found that the induced emf across the loop is 125 mV. The angle θ is: **[1]**

a) 45°

b) 60°

c) 90°

d) 30°

8. The magnetic moment of a revolving electron around the nucleus varies with principal quantum number n as **[1]**

a) $\mu \propto \frac{1}{n}$

b) $\mu \propto n$

c) $\mu \propto n^2$

d) $\mu \propto \frac{1}{n^2}$

9. In a diffraction pattern due to a single slit of width a, the first minimum is observed at an angle 30° when light of wavelength 5000 $\mathring{A}$ is incident on the slit. The first secondary maximum is observed at an angle of **[1]**

a) $\sin^{-1}\left(\frac{1}{2}\right)$

b) $\sin^{-1}\left(\frac{1}{4}\right)$

c) $\sin^{-1}\left(\frac{3}{4}\right)$

d) $\sin^{-1}\left(\frac{2}{3}\right)$

10. The magnitude of electric force, F is: **[1]**

a) directly proportional to the product of both charges.

b) constant

c) directly proportional to the distance between both charges

d) directly proportional to the square of the distance between both charges.

11. In the given Fig., V_o is the potential barrier across a p-n junction, when no battery is connected across the junction **[1]**

a) 3 and 1 both correspond to reverse bias of junction.

b) 3 corresponds to forward bias of junction and 1 corresponds to reverse bias of junction

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c) 1 corresponds to forward bias and 3 corresponds to reverse bias of junction.

d) 1 and 3 both correspond to forward bias of junction

12. Which of the following is used in optical fibers? **[1]**

a) Refraction

b) Scattering

c) Total internal reflection

d) Diffraction

13. **Assertion (A):** If the frequency of the incident light on a metal surface is doubled, the kinetic energy of emitted electrons is more than doubled. **[1]**

Reason (R): The metal will provide additional energy to the emitted photoelectron for light of higher frequency than that for lower frequency.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion (A):** The potential of earth is assumed zero. **[1]**

Reason (R): Earth is insulator and so earth can not hold any charge.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

15. **Assertion (A):** When a light wave travels from rarer to denser medium, its speed decreases. Due to this reduction of speed the energy carried by the light wave reduces. **[1]**

Reason (R): Energy of wave is proportional to the frequency.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

16. A resistor, a capacitor and an inductor are connected in series. The combination is connected across an ac source of frequency 50 Hz. **[1]**

Assertion: Peak current through each remains same.

Reason: Average power delivered by source is equal to average power developed across resistance.

a) Assertion and reason both are correct statements and reason is correct explanation for assertion.

b) Assertion is correct statement but reason is wrong statement.

C) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

d)Assertion is wrong statement but reason is correct statement.

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